

# CRISISGUARD AI

Unified Multi-Hazard Early Warning, Real-Time Geospatial Intelligence & Pilgrimage Evacuation Architecture

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|------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Target Region: Uttarakhand Himalayas (Char Dham Corridors: NH-07 & NH-107)                     | Version: 2.0 (SIH Evaluation Release)         |
| Official Nodal Alignment: IMD (Ministry of Earth Sciences) & GSI BhuSanket (Ministry of Mines) | System Status: Operational / Fully Integrated |

## 1. Executive Summary & Problem Scope

Every pilgrimage season, over 4.5 million devotees traverse Uttarakhand's fragile mountain highways (Kedarnath, Badrinath, Gangotri, Yamunotri). The region is severely susceptible to compounding, high-velocity natural hazards: catastrophic rainfall-induced **landslides**, localized orographic **cloudbursts**, **Glacial Lake Outburst Floods (GLOF)**, severe **crowd bottleneck panics** on narrow trek paths, and viral **social media crisis disinformation**. Existing institutional systems operate in silos — weather alerts remain separate from landslide slope models, pilgrim tracking is detached from hazard maps, and rumor debunking takes hours.

**CrisisGuard AI** solves this through a unified, real-time spatial platform that continuously correlates live **IMD Doppler Weather Radar** (250km sweep from Dehradun), **GSI BhuSanket Landslide Susceptibility** models, high-altitude Automatic Weather Stations (AWS), and pilgrim mobility density into a single operational picture for civilians, responders, and state administrators.

## 2. End-to-End 4-Tier System Topology

The CrisisGuard AI architecture is divided into 4 tightly decoupled, highly resilient tiers:

| Tier Layer                                | Core Components & Integration Engines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Key Output / Responsibilities                                                                                                            |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1:<br>Data Ingestion & Telemetry     | <ul style="list-style-type: none"><li>• <b>IMD Gateway API (api.imd.gov.in)</b>: Doppler Radar (/radar_image), 3-Hr District Nowcast (/nowcast_district), AWS (/aws_data), NHAI Highway alerts.</li><li>• <b>GSI BhuSanket LEWS</b>: Landslide susceptibility image server &amp; landslide inventory REST servers.</li><li>• <b>High-Altitude Sensor Nodes</b>: Soil moisture probes, rainfall tip buckets, river level gauges.</li><li>• <b>Crowdsourced Field Reports</b>: Civilians &amp; SDRF geo-tagged incident logs.</li></ul>                                                    | Sub-second ingestion of meteorological, geological, and crowd density telemetry; normalized into unified GeoJSON standards.              |
| Tier 2:<br>AI Intelligence & Risk Engines | <ul style="list-style-type: none"><li>• <b>Slope Stability &amp; Landslide Engine</b>: Antecedent rainfall saturation + slope gradient factoring.</li><li>• <b>Convective Cloudburst Engine</b>: Doppler reflectivity thresholding (&gt;45 dBZ, rain rate &gt;50 mm/hr).</li><li>• <b>GLOF Hydrodynamic Model</b>: Moraine lake expansion rate + downstream flood transit time.</li><li>• <b>Crowd Chokepoint Engine</b>: Pilgrim density per m² + queue wait times.</li><li>• <b>Fake News Buster (NLP)</b>: Cross-references social claims against USDMA/SDRF incident logs.</li></ul> | Computes composite Threat Scores (0-100), automated 4-tier alert level transitions (Green/Yellow/Orange/Red), and evacuation lead times. |
| Tier 3:<br>Geospatial GIS Engine          | <ul style="list-style-type: none"><li>• <b>React-Leaflet 4.2 &amp; Leaflet 1.9 Engine</b>: High-performance interactive vector mapping.</li><li>• <b>Spatial Buffer Overlays</b>: Dynamic threat rings (2.5 km to 22 km buffer zones).</li><li>• <b>Corridor Route Polylines</b>: NH-07 (Rishikesh-Badrinath) and NH-107 (Rudraprayag-Kedarnath).</li><li>• <b>Layer Controller</b>: Independent toggle switches for Landslides, Cloudbursts, GLOF, Crowd, Alerts, and official IMD/GSI feeds.</li></ul>                                                                                 | Interactive visualization of real-time disaster footprints, active convoys, and safe evacuation corridors.                               |

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|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Tier 4:<br>Omnichannel<br>Delivery &<br>Portals | <ul style="list-style-type: none"><li>• <b>Pilgrim Mobile Web / PWA:</b> GPS Journey Risk Calculator, Safe Shelter locator, SOS trigger.</li><li>• <b>Field Responder PWA:</b> SDRF / NDRF incident verification &amp; resource dispatch logs.</li><li>• <b>State Command Center:</b> District-level multi-hazard heatmaps, emergency broadcast triggers.</li><li>• <b>SMS / Low-Bandwidth Fallback:</b> Critical alerts delivered without internet via telecom Cell Broadcast.</li></ul> | Tailored, persona-specific interfaces ensuring right information reaches the right stakeholder at the right second. |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

3. Detailed Blueprint of the 5 Core Hazard Engines

| Engine Name                  | Trigger Criteria & Science                                                                                                                                         | Data Sources & Nodal Links                                                                                                                                                       | Operational Output & Mitigation                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1. Landslide Risk Engine     | Calculates Factor of Safety (FoS) based on cumulative 72-hr rainfall saturation, soil moisture %, slope inclination (>35°), and active seismic/blasting triggers.  | <ul style="list-style-type: none"><li>• GSI BhuSanket (LEWS)</li><li>• IMD Rainfall Thresholds</li><li>• Rudraprayag Pilot Polygon</li></ul>                                     | Km-wise highway hazard categorization on NH-07 (Lambagar chute) & NH-107. Triggers SDRF convoy escort passage.               |
| 2. Cloudburst Nowcast Engine | Detects rapid vertical cloud development and extreme convective precipitation (>50 mm/hr in under 60 mins) within narrow orographic catchments.                    | <ul style="list-style-type: none"><li>• IMD Dehradun Doppler Radar (/radar_image)</li><li>• IMD District Nowcast (/nowcast_district)</li><li>• Kedarnath AWS rain rate</li></ul> | 3-hour localized advance nowcasting window. Automatic triggering of riverside camp sirens and trek suspension.               |
| 3. GLOF Glacial Lake Engine  | Monitors proglacial and supraglacial lake surface area expansion, moraine dam freeboard reduction, and seismic/ice-avalanche impact displacement.                  | <ul style="list-style-type: none"><li>• Sentinel-2 Multispectral SAR</li><li>• Wadia Institute (WIHG) archives</li><li>• Downstream telemetry gauges</li></ul>                   | Downstream flood transit calculations: Sonprayag (45 min lead time), Gaurikund (18 min). Automatic holding camp evacuations. |
| 4. Crowd Chokepoint Engine   | Monitors pedestrian density exceeding 4 persons/m² on narrow footpaths, mule staging queues, and transit hub bottlenecks (Sonprayag, Gaurikund).                   | <ul style="list-style-type: none"><li>• UTDB RFID Pilgrim Yatra Cards</li><li>• Mobile tower density telemetry</li><li>• Drone / CCTV count feeds</li></ul>                      | Regulated batch release from Guptkashi holding camps. Real-time queue wait time alerts (e.g. 4.5h wait at Gaurikund).        |
| 5. Fake News Buster Engine   | NLP-based social claim verification against official USDMA, SDRF, and IMD incident logs to detect sensationalized, recycled crisis videos & false collapse rumors. | <ul style="list-style-type: none"><li>• USDMA Control Room Log</li><li>• SDRF Field Dispatch Log</li><li>• Social Media API monitors</li></ul>                                   | Official Credibility Score (0-100%) and instant public debunking advisories to prevent mass stampedes and panic evacuations. |

4. Geospatial GIS & Official Government Data Gateway

CrisisGuard integrates directly with the official Government of India repositories as documented in the **IMD API Reference** ([https://api.imd.gov.in/public/api\\_reference.html](https://api.imd.gov.in/public/api_reference.html)) and **GSI BhuSanket LEWS** (<https://bhusanket.gsi.gov.in>).

| Government Gateway                  | Mapped Endpoint          | Parameters & Auth                                  | Map Rendering in CrisisGuard                                                                                                                 |
|-------------------------------------|--------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| IMD (MoES)<br>District Warnings     | /api/v1/district_warning | state=UTTARAKHAND<br>Header: x-api-key             | Interactive polygon circles color-coded by IMD 4-tier alert system (Red/Orange/Yellow/Green) across 13 Uttarakhand districts.                |
| IMD (MoES)<br>Doppler Weather Radar | /api/v1/radar_image      | radar_name=DEHRADUN<br>Band: C-Band Polarimetric   | 250km radar sweep range circle centered at Dehradun (30.34°N, 78.04°E) + convective reflectivity echo clusters (e.g. Mandakini 52 dBZ cell). |
| IMD (MoES)<br>NHAI Highway Nowcast  | /api/v1/highway_nowcast  | Corridor: NH-07 & NH-107<br>Validity: 3 to 6 Hours | Highway hazard pins along Char Dham routes: Km 282 Lambagar Chute (NH-07) and Sonprayag-Gaurikund axis (NH-107).                             |

|                                          |                                         |                                                                 |                                                                                                                             |
|------------------------------------------|-----------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| IMD (MoES)<br>High-Altitude AWS          | /api/v1/aws_data                        | Stations: Kedarnath, Badrinath, Joshimath, Sonprayag            | Telemetry pins displaying real-time temperature, hourly rainfall (mm/h), relative humidity %, and barometric pressure.      |
| GSI BhuSanket<br>Landslide Early Warning | /gisserver/rest/services/Hosted/LFS_map | Pilot District: <b>Rudraprayag</b><br>ArcGIS REST FeatureServer | Historical landslide polygons, slope susceptibility heatmaps, and active rockfall alerts for Kedarnath corridor.            |
| Vite Dev Proxy<br>CORS Resolution        | /api/imd/* →<br>api.imd.gov.in/api/v1/* | Configured in vite.config.js<br>changeOrigin: true              | Bypasses browser CORS policy restrictions in client development; automatic fallback to cached high-fidelity offline schema. |

5. Frontend Codebase & Component Tree

Built on modern **React 18**, **Vite**, and **Tailwind CSS** with full responsive adaptation from mobile phones to command center displays:

| Directory / Module     | File Path                                                                    | Purpose & Role in CrisisGuard                                                                                                                                 |
|------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GIS Map Component      | src/components/map/LeafletRiskMap.jsx                                        | Standalone interactive Leaflet map featuring OpenStreetMap tiles, Char Dham polylines, IMD Doppler Radar overlay, district warnings, and attribution ribbons. |
| Full Risk Map Page     | src/pages/RiskMap/RiskMap.jsx                                                | Full-screen situational page with sector presets (Kedarnath, Badrinath), layer control flyout, IMD banner, and interactive IMD API Guide Modal.               |
| Map Controls & Legend  | src/components/map/RiskLayerControl.jsx<br>src/components/map/RiskLegend.jsx | Floating layer manager with granular hazard switches + IMD Telemetry toggles. Floating legend explaining 4-tier alert colors and radar reflectivity.          |
| Official IMD Service   | src/services/imdService.js                                                   | Encapsulates all IMD and GSI BhuSanket API endpoints, auth token handling, and realistic high-altitude offline telemetry fallback.                            |
| Journey Risk Planner   | src/pages/JourneyPlanner/                                                    | Origin-Destination route risk calculator with separated boarding point selectors, weather forecast along route, and safe shelter recommendations.             |
| Safe Shelters & Relief | src/pages/SafeShelters/                                                      | Evacuation camp directory showing free bed counts, medical post availability, food supplies, and SOS contact numbers.                                         |
| Fake News Verifier     | src/pages/VerifyNews/                                                        | NLP rumor debunking engine allowing users to paste WhatsApp/social media claims and receive verified official bulletins.                                      |

6. Official Emergency Action Protocol Matrix

Standardized in accordance with National Disaster Management Authority (NDMA) and IMD color-coded warning scales:

| Alert Level | IMD Color Code        | Scientific Hazard Conditions                                                                                   | Enacted Operational Mandate                                                                                            |
|-------------|-----------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| GREEN       | No Warning            | Rainfall <15 mm/day, soil saturation <40%, clear radar echoes, crowd capacity <70%.                            | Normal pilgrimage transit. Standard SDRF highway patrol operating.                                                     |
| YELLOW      | Be Updated (Watch)    | Rainfall 15-64 mm/day, gusty winds 30-40 km/h, radar echoes 30-40 dBZ, crowd capacity 70-90%.                  | Advisory broadcast to pilgrim apps. Staging earthmovers near Lambagar & Sirobagarh.                                    |
| ORANGE      | Be Prepared (Alert)   | Very heavy rain (65-115 mm/day), high landslide vulnerability, radar echoes 40-48 dBZ, crowd capacity 90-110%. | Single-lane escorted convoys on NH-07/NH-107. Night transit halted. Regulated batch release at Sonprayag.              |
| RED         | Take Action (Warning) | Extremely heavy rain (>115 mm/day), cloudburst detected (>50 mm/h, >48 dBZ), GLOF watch, crowd overload >110%. | Immediate suspension of yatra. Evacuation of riverside camps to designated safe shelters. Full SDRF/NDRF mobilization. |

## 7. Innovation Highlights for SIH Evaluation

Why CrisisGuard AI stands distinct from existing disaster portals:

- **Dual-Agency Real-Time Synchronization:** First platform to synchronously correlate atmospheric triggers from **IMD (Doppler Radar)** with geological slope models from **GSI BhuSanket** for the Char Dham corridor.
- **High-Altitude Offline-First Architecture:** When mountain connectivity drops in remote gorges, CrisisGuard's local PWA cache ensures continuous situational guidance and safe shelter routing.
- **Crowd Chokepoint & Mule Staging Balancing:** Prevents stampedes by synchronizing transit release gates at Guptkashi with real-time trek capacity at Gaurikund.
- **Misinformation Immunization:** Direct NLP verification against official SDRF logs eliminates dangerous panic-inducing rumors during crisis events.
- **Production-Ready Codebase:** Full React 18 / Tailwind CSS implementation, passing production builds with zero errors, and clean reverse-proxy gateway routing.

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CrisisGuard AI • Built with dedication for the safety of Himalayan pilgrims and disaster response forces.